



Proposed CEA Link — December 2026

The December 2026 timetable rules with no duty over 8h40 and the ticket office written in — two 14:00–22:30 turns a day Monday to Saturday, two 13:30–22:00 on a Sunday — judged on the ORR fatigue factors

Prepared 22 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylbone Roster v24.22) · figures on this page are computed, not typed

PROPOSAL

By the Book 2

Eight Forty, with the ticket office rostered — two 14:00–22:30 a day

CODE **B2-24-G21**

FINGERPRINT **02f3c005**

BUILT FROM duty table G · seed 21 · 24 lines · 4 cover weeks

LINEAGE Family B2 — “Eight Forty” (EF-24-E21 · 0cf19f56) with the ticket office written in: two 14:00–22:30 turns every day Monday to Saturday and two 13:30–22:00 on a Sunday, fixed before the rest of the day was searched. Same 8h40 ceiling, same December 2026 timetable rules, demand fit ahead of the count of distinct times (owner, 22 Sep 2026). The rotation was then searched for the ORR factors exactly as “By the Book” and “Eight Forty” were. Compare with “Eight Forty”, “By the Book” (BB-24-D7 · 0f14abce) and “Same Turns” (ST-24-B7 · d15e1b74).

✓ 24 lines designed

✓ 35h 00m a week — exactly the contract

✓ 0 hard-limit breaches

✓ 0 rests under 12h

✓ 0 fatigue factors present (today: 5)

26 turns

in the December 2026 duty table, capped

searched against the timetable under the 8h40 rule; later shorter than earlier; the longest duty is 8h 40m

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

0 of 25

fatigue factors present

today's link has 5; every other factor is clear or does not apply

5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

2h 11m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 — *By the Book* under one more rule: **no duty runs over 8h40**. That rule is not a trim. Five of *By the Book*'s eleven Monday-to-Saturday turns ran past it (its earlies reached 9h30), and a day's minutes are fixed by the contract — 14 duties paying 7,000 minutes is a mean of 8h20 — so with the ceiling twenty minutes above the mean, every duty has to sit between 7h and 8h40 and most of them near the top. The December 2026 duty table was therefore searched again from nothing under the same rules: four to open every day, three through to the close and four on a Saturday, five still on at 22:00, fourteen on a Saturday and ten on a Sunday, late turns shorter than early ones, no :05 or :10 times, at most two people on a turn, and the times set against the measured December 2026 service. It pays the 35-hour week to the minute across 20 working lines and four cover weeks.

The rotation was then searched for one thing: the rules it will be assessed against. Chiltern's 13-day limit, twelve hours between duties and the exact contract were treated as walls; the 25 ORR good-practice fatigue factors were the objective, each one scored by the workspace's own `assessFatigue` on every one of about a million candidate arrangements, together with the size of the week-to-week step, weekends off and an even spread of cover weeks. Only after every factor was settled was a week's coherence allowed to count — one turn a week where it costs nothing against the factors — so that a rotation is not chopped up for no fatigue gain. The rules-only version, with no such term, is on page 8 for comparison. Page 8 says how, and how to reproduce it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is
- 2 The decision it supports (this page)
- 3 The 24-line grid, with the words explained
- 4 A week compared with today
- 5 People on duty, hour by hour
- 6 Hard limits and the December 2026 figures
- 7 The ORR fatigue factors
- 8 Where it came from
- 9 The grid, ready to paste into the app

The question

Should the link for the December 2026 timetable be built from the December 2026 timetable rules and the measured timetable, accepting shift times nobody works today, or keep today's times?

By the Book 2 is a rules-first table with the ticket office rostered. It is read against Quarter To (QT-24-Q34), the strongest case for keeping today's times — not against every design in the folder.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 50m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **3 of 4** (today 0 of 4)
- Fatigue factors present: **0** (today 5)
- Demand fit weekday · Sat · Sun: **67.9 · 8.1 · 35.8** (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **4 of 26** · full weekends off: **5** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 00m–8h 40m, lates 7h 05m–8h 40m
- 22 of 26 clock times are new — the duty table on page 4 lists them
- The longest duty, 8h 40m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 14/10 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 6 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
Four at the open, three to close, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
0 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 67.9 · 8.1 · 35.8	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
4 of 26 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability
Search score 19060	The annealer's own objective; page 8 explains it	Do not weigh

What no page here can tell you

- Whether the people who would work them accept a table in which 22 of 26 times are new (page 4 lists them) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (11 as a block, 12 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 2, 3, 4, 5, 6 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	16:00 23:25	14:00 22:30	RD	RD	14:00 22:30	14:00 22:30	15:50 23:55	33h 35m	41h 00m	4	1
3	15:20 22:25	15:50 23:55	15:50 23:55	RD	RD	15:50 23:55	14:00 22:30	32h 45m	39h 50m	4	1
4	RD	15:45 23:55	15:45 23:55	14:00 22:30	14:00 22:30	RD	14:00 22:30	41h 50m	41h 50m	5	2
5	15:20 23:25	14:00 22:30	14:00 22:30	14:00 22:30	RD	RD	06:20 15:00	34h 10m	42h 15m	4	1
6	07:15 15:55	06:20 14:50	06:20 14:50	RD	RD	06:20 14:50	07:45 16:25	34h 10m	42h 50m	4	1
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	15:55 23:25	RD	RD	16:20 23:55	16:20 23:55	16:20 23:55	15:00 22:50	30h 35m	38h 05m	4	1
9	13:30 22:00	RD	RD	15:45 23:55	15:45 23:55	15:45 23:55	RD	24h 30m	33h 00m	3	1
10	RD	10:55 19:35	10:55 19:35	10:55 19:35	10:55 19:35	10:55 19:35	RD	43h 20m	43h 20m	5	1
11	RD	10:55 19:35	10:55 19:35	10:55 19:35	10:55 19:35	10:55 19:35	RD	43h 20m	43h 20m	5	1
12	RD	16:20 23:55	16:20 23:55	RD	RD	14:00 22:00	15:40 23:55	31h 25m	31h 25m	4	2
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
14	07:15 15:50	09:15 17:55	RD	09:15 17:55	09:15 17:55	09:15 17:55	RD	34h 40m	43h 15m	4	1
15	RD	09:15 17:55	09:15 17:55	RD	06:20 14:55	06:20 14:55	06:20 13:55	42h 05m	42h 05m	5	2
16	RD	RD	RD	06:20 15:00	06:20 15:00	06:20 15:00	06:20 14:55	34h 35m	34h 35m	4	1
17	RD	RD	14:00 22:30	15:50 23:55	15:50 23:55	14:00 22:30	15:45 23:55	41h 20m	41h 20m	5	2
18	RD	RD	14:00 22:00	14:00 22:00	14:00 22:00	RD	15:55 23:55	32h 00m	32h 00m	4	1
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
20	13:30 22:00	14:00 22:00	RD	06:20 14:50	06:20 14:50	RD	RD	25h 00m	33h 30m	3	2
21	RD	RD	09:15 17:55	09:15 17:55	09:15 17:55	09:15 17:55	07:45 16:25	43h 20m	43h 20m	5	1
22	RD	06:20 14:55	06:20 14:55	06:20 14:55	RD	RD	06:20 14:50	34h 15m	34h 15m	4	1
23	07:15 15:35	06:20 15:00	06:20 15:00	06:20 13:45	06:20 13:45	RD	RD	32h 10m	40h 30m	4	2
24	07:15 14:15	06:20 13:45	06:20 13:45	RD	RD	06:20 13:45	07:20 16:00	30h 55m	37h 55m	4	1
Cover	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 19m	4.20	11 one turn
Minutes	4,780	7,000	7,000	7,000	7,000	7,000	7,000	42,000	46,780		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

How a week compares with today

Nothing here was asked to resemble today's roster; this is how close the fatigue factors took it on their own, and where the December 2026 duty table differs

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	11 of 20
Weeks mixing early and late turns	7	5
Rest-day breaks that are two days or more	13 of 17	18 of 25
Single rest days between duties (not a 48h break)	4	7
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Distinct shift times (4 on today's roster)	18	26
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	14	owner's figure for December 2026, leaning late for events
Sunday	8	10	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	4 · 4 · 4	four to open, every day — met
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	3 · 4 · 3	three to close, four on a Saturday — met
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	5 · 7 · 4	five from 22:00, Monday to Saturday — not met: Saturday 7

The duty table — the December 2026 timetable rules under 8h40, beside today's

DECEMBER 2026 TURN	WK	SAT	SUN
06:20–13:45 7h 25m · Early	1		
06:20–13:55 7h 35m · Early		1	
06:20–14:50 8h 30m · Early	1	1	
06:20–14:55 8h 35m · Early	1	1	
06:20–15:00 8h 40m · Early	1	1	
07:15–14:15 7h 00m · Early			1
07:15–15:35 8h 20m · Early			1
07:15–15:50 8h 35m · Early			1
07:15–15:55 8h 40m · Early			1
07:20–16:00 8h 40m · Early		1	
07:45–16:25 8h 40m · Early		2	
09:15–17:55 8h 40m · Early	2		
10:55–19:35 8h 40m · Early	2		
13:30–22:00 8h 30m · Late			2
14:00–22:00 8h 00m · Late	1		
14:00–22:30 8h 30m · Late	2	2	
15:00–22:50 7h 50m · Late		1	
15:20–22:25 7h 05m · Late			1
15:20–23:25 8h 05m · Late			1
15:40–23:55 8h 15m · Late		1	
15:45–23:55 8h 10m · Late	1	1	
15:50–23:55 8h 05m · Late	1	1	
15:55–23:25 7h 30m · Late			1
15:55–23:55 8h 00m · Late		1	
16:00–23:25 7h 25m · Late			1
16:20–23:55 7h 35m · Late	1		

Read the table in three blocks: 6 turns Monday to Saturday alike (and 5 the weekday keeps to itself), Saturday's own 6, and Sunday's 9 for its 07:15–23:25 window. Openers stagger by their finish and closers by their start so nobody hands over a cliff. 15 of the 35 start and finish instances sit on :00, :15, :30 or :45; the 16 that do not (07:20, 10:55, 13:55, 14:50, 14:55, 15:20, 15:35, 15:40, 15:50, 15:55, 16:20, 16:25, 17:55, 19:35, 22:25, 22:50) are what four distinct opener finishes cost under a cap that leaves the long earlies about twenty minutes of room — *By the Book* has three such times.

TODAY'S TURN (NONE IS REUSED)	WK	SAT	SUN
06:20–13:35 7h 15m	1		
06:20–13:45 7h 25m	1		
06:20–14:00 7h 40m		1	
06:20–14:20 8h 00m	2		
06:20–14:50 8h 30m		3	
07:15–15:45 8h 30m			3
08:00–16:30 8h 30m	2		
08:30–16:30 8h 00m			1
11:00–19:30 8h 30m	1		
12:00–20:00 8h 00m		1	
13:00–21:00 8h 00m			1
13:30–21:00 7h 30m		1	
13:30–22:00 8h 30m	1		
14:00–22:30 8h 30m	2		
14:30–22:00 7h 30m		2	
14:30–23:25 8h 55m			3
14:45–23:55 9h 10m		2	
15:15–23:55 8h 40m	2		

06:20–13:45 is the one short early; every other early (8h 30m–8h 40m) is longer than every late (7h 35m–8h 30m) — the lever against unpopular lates, at the only size the cap allows: the shortest long early is 0 minutes longer than the longest late, where *By the Book* had fifteen and an hour on average. 4 of the 26 turns are times people work today (06:20–13:45, 06:20–14:50, 13:30–22:00, 14:00–22:30) — the same trade as *By the Book*, and the question to put to the room rather than to the tool.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED		4	4	4	6	8	8	8	8	10	9	10	10	8	8	6	6	5	3	67.9
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		4	7	7	7	7	7	7	7	8	10	9	7	7	7	7	7	7	4	8.1
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			4	4	4	4	4	4	6	6	8	6	6	6	6	6	6	4	3	35.8
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

Reading it. The table's times were searched against this curve on a five-minute grid, so the cover follows the traffic as closely as the day's fixed minutes allow. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **13h 50m** — **line 21 Fri 09:15-17:55 into Sat 07:45-16:25** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (79.67h) sit on top as RDW, as they do today. Individual weeks range 24h 30m–43h 20m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	✓ 4 weekday · 4 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	✓ 3 weekday · 4 Saturday · 3 Sunday	
Five still on duty at 22:00	× 5 weekday · 7 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	✓ 14 · 10	
Four cover weeks, evenly spread	✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6	whole weeks, never scattered days
About 4.2 days a week worked, Mon-Sat	✓ 4.20 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	today's times already obey this
Late turns slightly shorter than most earlies	× weekday not met (lates 7h35–8h30, earlies 7h25–8h40) · Saturday not met (lates 7h50–8h30, earlies 7h35–8h40) · Sunday not met (lates 7h05–8h30, earlies 7h00–8h40)	incompatible with keeping today's times: meeting it needs 9h+ earlies or 15 duties a day — flagged for decision, below
No duty over 8h40	✓ longest 8h40	every day class — By the Book's longest is 9h30, today's 9h10

To settle before it is frozen

The ticket office is written in, and it changed what the rules could give. Two 14:00–22:30 turns Monday to Saturday and two 13:30–22:00 on a Sunday were fixed before anything else was searched (owner, 22 Sep 2026). They sit outside the early-versus-late ordering rule, which an 8h30 late would otherwise make unsatisfiable under an 8h40 ceiling. **Two pins are read differently for them, and only for them:** "five still on at 22:00" is a floor (four Saturday closers plus the pair make six by arithmetic, and the rule exists so the evening is not thin), and a pinned turn finishing AT 22:00 counts as on at 22:00, while design duties keep the strict reading, so *Eight Forty's* Saturday assesses as before. **Fit before count** (owner): 67.9 weekday, 8.1 Saturday, 35.8 Sunday against *Eight Forty's* 70.4, 16.4 and 20.9. Saturday is the best table in the folder; Sunday is the price — the pair sits across its quietest afternoon, and Sunday pays 4,780 minutes against 4,820. **Familiarity.** 4 of 26 turns are times people work today (06:20–13:45, 06:20–14:50, 13:30–22:00, 14:00–22:30). **Sunday's finish** — five December 2026 timetable movements fall after 23:25; the window is inherited, not decided.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). $\triangle$ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. Nathan assesses against this list.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 24
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	$\triangle$ 3	$\checkmark$ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 12 $\checkmark$ 11 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	$\triangle$ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	$\triangle$ 60.8	$\checkmark$ 51.9
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	$\triangle$ 9	$\checkmark$ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 12 backward / 12 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 2h 11m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	$\triangle$ 6	$\checkmark$ 0
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 0 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was chosen

Reproducible: the same table, seed and objective give the same grid on any machine

In plain terms. A computer took each day's duties as fixed and tried over a million different ways of arranging them across the 24 weeks, scoring every arrangement on the rules on pages 6 and 7, and kept the best. The "seed" is the number that makes that run repeat exactly on any machine, so anyone can rebuild this grid and get the same cells.

How it was chosen

1 • The table. The December 2026 timetable rules under the 8h40 cap, with two ticket-office turns fixed first — 14:00–22:30 twice a day Monday to Saturday, 13:30–22:00 twice on a Sunday — and the rest of each day searched around them. The table was not annealed: table-book.mjs's own search walks lengths and starts together and, with two duties pinned, reports no feasible table where one exists. So every legal set of duty LENGTHS was enumerated with the pinned pair taken out (PIN_N/PIN_MIN), and each one was placed (place-structures.mjs) — 139,017 weekday structures, 12 feasible; 18,616 Saturday, 931; 24,942 Sunday at its total, 3601. The pick was demand fit first, then the count of distinct times (owner). The pinned pair sits outside the early-versus-late ordering rule and counts as on at 22:00 when it finishes there; page 6 says why, and every other rule counts it.

2 • The grid. Simulated annealing over the 24-line grid: 100,000 candidate arrangements per restart, five restarts from different random starting points per run, four seeded runs — over a million candidates in all — then a steepest-descent polish of every same-

day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

3 • The judge. Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, then every fatigue factor present (each one, and its size: the length of an early block, the hours in seven days, each start-time jump, each backward step), then the week-to-week step and the length of a block on the same sort of week. Then, at a weight no factor can be traded for, the coherence of a week (one turn, its weekend variant) and fairness: four to five days a week, no single rest day between duties. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
B2-24-G21 · 02f3c005 — By the Book 2 (this proposal)	6	0	5	11/20	2h 11m	67.9	19060
B2-24-G13 · cf3b1372 — runner-up	6	1	6	11/20	1h 41m	67.9	62810
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	2192436
EF-24-E21 · 0cf19f56 — Eight Forty (a sibling proposal)	6	0	6	10/20	1h 31m	70.4	—
BB-24-D7 · 0f14abce — By the Book (a sibling proposal)	6	0	6	12/20	2h 52m	30.8	—
ST-24-B7 · d15e1b74 — Same Turns (a sibling proposal)	6	1	6	18/20	2h 33m	46.3	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	44.7	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores). fatigue factors present decided it (0 against 1 for the runner-up). The other 2 seeded results are in results/ (G7, G34); none stands higher on the pick than the rows shown. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as By-the-Book-2-B2-24-G21-import.txt (tab-separated, pastes directly) and By-the-Book-2-B2-24-G21.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	16:00-23:25	14:00-22:30	RD	RD	14:00-22:30	14:00-22:30	15:50-23:55
3	15:20-22:25	15:50-23:55	15:50-23:55	RD	RD	15:50-23:55	14:00-22:30
4	RD	15:45-23:55	15:45-23:55	14:00-22:30	14:00-22:30	RD	14:00-22:30
5	15:20-23:25	14:00-22:30	14:00-22:30	14:00-22:30	RD	RD	06:20-15:00
6	07:15-15:55	06:20-14:50	06:20-14:50	RD	RD	06:20-14:50	07:45-16:25
7	SP	SP	SP	SP	SP	SP	SP
8	15:55-23:25	RD	RD	16:20-23:55	16:20-23:55	16:20-23:55	15:00-22:50
9	13:30-22:00	RD	RD	15:45-23:55	15:45-23:55	15:45-23:55	RD
10	RD	10:55-19:35	10:55-19:35	10:55-19:35	10:55-19:35	10:55-19:35	RD
11	RD	10:55-19:35	10:55-19:35	10:55-19:35	10:55-19:35	10:55-19:35	RD
12	RD	16:20-23:55	16:20-23:55	RD	RD	14:00-22:00	15:40-23:55
13	SP	SP	SP	SP	SP	SP	SP
14	07:15-15:50	09:15-17:55	RD	09:15-17:55	09:15-17:55	09:15-17:55	RD
15	RD	09:15-17:55	09:15-17:55	RD	06:20-14:55	06:20-14:55	06:20-13:55
16	RD	RD	RD	06:20-15:00	06:20-15:00	06:20-15:00	06:20-14:55
17	RD	RD	14:00-22:30	15:50-23:55	15:50-23:55	14:00-22:30	15:45-23:55
18	RD	RD	14:00-22:00	14:00-22:00	14:00-22:00	RD	15:55-23:55
19	SP	SP	SP	SP	SP	SP	SP
20	13:30-22:00	14:00-22:00	RD	06:20-14:50	06:20-14:50	RD	RD
21	RD	RD	09:15-17:55	09:15-17:55	09:15-17:55	09:15-17:55	07:45-16:25
22	RD	06:20-14:55	06:20-14:55	06:20-14:55	RD	RD	06:20-14:50
23	07:15-15:35	06:20-15:00	06:20-15:00	06:20-13:45	06:20-13:45	RD	RD
24	07:15-14:15	06:20-13:45	06:20-13:45	RD	RD	06:20-13:45	07:20-16:00

Prepared 22 September 2026 · re-rendered 24 September 2026. The code **02f3c005** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.